From Free–Energy Loops to Human Values: Hubs as Bottlenecks

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Abstract

Zarncke [2025] catalogued fifteen free–energy–minimising loops in the vertebrate brain. Each loop closes a prediction–error feedback and projects into one or more neuromodulatory or salience "hubs" that broadcast precision weights. Here we introduce the *loop–hub–value* (LHV) model: an architectural proposal in which a small set of hubs transforms high–dimensional error vectors into low–bandwidth scalars that are then read out as human value tags. We (i) formalise the two transfer steps (*loop* $\rightarrow$ *hub* and *hub* $\rightarrow$ *value*); (ii) compile an empirical mapping between hubs and intrinsic values, drawing on Zarncke [1]; and (iii) outline falsifiable predictions. Details of agent implementation and learning dynamics will appear in Part III.

1 Motivation

Biological agents cannot transmit every prediction error upstream; long–range bandwidth is constrained by wiring and metabolic costs. Evolution appears to have solved this with *hub bottlenecks*: mid–brain nuclei and limbic structures that *compress* multidimensional errors into one or two slow neuromodulatory signals. Cortex, basal ganglia and cerebellum decode those signals as *values* that steer policy selection.

2 Formal Skeleton of the LHV Model

Error compression. Each loop i produces an error vector $\boldsymbol{\epsilon}_i(t) \in \mathbb{R}^{d_i}$. A hub h receives a weighted sum over incoming loops

$$s_h(t) = \sigma\Big(\sum_{i \in \mathcal{I}(h)} w_{ih} \, \|\boldsymbol{\epsilon}_i(t)\|_p\Big), \tag{1}$$

where σ is a saturating non–linearity and w_{ih} are learned transfer weights.

Value tagging. A cortical decoder D maps hub signals to symbolic values $\mathcal{V} = \{v_1, \ldots, v_m\}$ via

$$P(v_k \mid \{s_h\}_h) = \mathrm{softmax}_k(\beta_k + \sum_h \alpha_{kh} \, s_h), \tag{2}$$

with α_{kh} contextually gated by attention and task set.

3 Empirical Hub–Value Map

Table 1 lists the hubs, the principal loops that feed them, and representative intrinsic values from Zarncke [1].

Hub	Dominant upstream loops	Example intrinsic values
Periaqueductal Gray (PAG)	Threatpain; Pavlovian threat loop	Protection, Non–suffering, Freedom
Ventral Tegmental Area (VTA)	Curiosity; Valuation regret	Learning, Diversity, Legacy
Locus Coeruleus (LC)	Precisiontiming	Achievement
Nucleus Accumbens (NAcc)	Valuationagency	Happiness
Orbitofrontal Cortex (OFC)	Policy–regret valuation	Appraisal, Hedonics
Hypothalamus	Metabolic; Steroidal timing	Longevity, Caring
HPG axis	Steroidal timing	Reputation
Insula	Metabolic; Morphological repair	Fairness, Purity
Anterior Cingulate Cortex (ACC)	Coalition entropy	Justice, Respect
Amygdala	Pavlovian threat tagging	Virtue
Septal Nuclei	Reproductive physiology	Loyalty
Superior Colliculus	Perimeter defence	Nature, Truth
Primary Visual Cortex	Scene novelty[†]	Beauty

Table 1: Hub–value correspondences. [†]The scene novelty loop is hypothesised and will be formalised in Part III.

4 Orphaned Hubs and Candidate Loops

Although most neuromodulatory hubs receive direct input from an identified free–energy loop, three sub-hubs in Table 1—*orbitofrontal cortex (OFC)*, *hypothalamo–pituitary–gonadal (HPG) axis*, and *amygdala*—currently lack a first-pass loop in the evolutionary ledger. We treat them as *bottleneck refinements*: second-stage processors that reshape upstream error signals onto slower or more specialised control channels. Below we sketch their provisional computations and outline candidate loops to be formalised in future work.

4.1 Orbitofrontal Cortex (OFC)

The OFC integrates multi-modal evidence with predicted outcomes to compute a *policy–regret* signal: the squared discrepancy

$$\mathbb{E}[(Q - Q)^2]$$

between the best attainable action value Q and the executed value Q. This counterfactual loop acts on a timescale of hundreds of milliseconds to seconds, refining valuation updates supplied by ventral striatum and VTA.

4.2 Hypothalamo–Pituitary–Gonadal (HPG) Axis

The HPG axis implements neuroendocrine control over fertility and life-history trade-offs. We posit a distinct *steroidal timing loop* that minimises a deviation term

$$|S_{\text{steroid}} - \hat{S}|,$$

where S_{steroid} denotes circulating gonadal hormones and $\hat{S}$ an energy- or age-adjusted set-point. Feedback operates over hours to months, providing a slow but powerful prior on reproductive strategy.

4.3 Amygdala

Beyond rapid nocifensive processing in the periaqueductal loop, the basolateral–central amygdala complex assigns socio-emotional salience. We hypothesise a *social appraisal loop* that reduces entropy over affective threat states,

$$H(\text{threat}_{\text{aff}}),$$

capturing mid-latency fear conditioning and norm-violation detection.

5 Predictions

1. Disrupting a hub (e.g., chemogenetically) should *simultaneously* perturb all values anchored to that hub, even if upstream loops remain intact.

2. Task contexts that enlarge $\|\boldsymbol{\epsilon}_i\|$ will modulate subjective value ratings in proportion to w_{ih}.

3. Agents endowed with an LHV bottleneck should yield more interpretable preference shifts than agents using flat reward summation when hub weights are perturbed.

6 Related Research

6.1 Loop–Through–Hub Circuitry

Weiller et al. [2] identify a dual–loop motif in human cortex where dorsal (sequencing) and ventral (conceptual) streams converge on frontal and posterior hubs, forming a closed circuit. Expanding this, Weiller et al. [3] describe a *meta–loop* in which lateral (external) and medial (internal) dual–loops interact via cross–hub projections, furnishing a substrate for integrating needs and demands.

Lyu et al. [4] provide causal evidence: effective–connectivity analyses reveal a recurrent loop between dorsal and ventral precuneus subdivisions that modulates DMN and fronto–parietal networks as tasks change. These findings validate the LHV claim that prediction–error messages funnel through hubs before influencing global policy.

6.2 Hub–Level Precision Weighting

Predictive–coding theory posits that precision (gain) of error signals is regulated by key hubs. Thalamic work by Kanai and Friston [5] highlights the pulvinar as a relay that encodes sensory reliability and synchronises relevant cortices, implementing precision control. Human fMRI by Hauser et al. [6] shows unsigned prediction–error signals in dorsal ACC and superior frontal cortex are precision–weighted in a dopamine–dependent manner, underscoring cortical hubs in error salience modulation.

6.3 Internal Value–External Information Integration

The salience network, centred on anterior insula and dorsal ACC, mediates switching between internal and external modes [7]. Malfunctions yield aberrant salience attribution, reinforcing its valuational gating role. Complementarily, fronto–striatal loops through ventral striatum and OFC relay dopaminergic reward–prediction signals, aligning instrumental value with ongoing cognition [8].

6.4 Error Compression Across Hierarchies

Predictive coding expects iterative suppression of redundant errors at each level. White–matter tractography reveals structural *bottlenecks* where converging fibres concentrate information flow [3].

Combined with neuromodulator–driven precision control, these bottlenecks plausibly compress the high–dimensional error space into the low–bandwidth hub scalars central to LHV.

References

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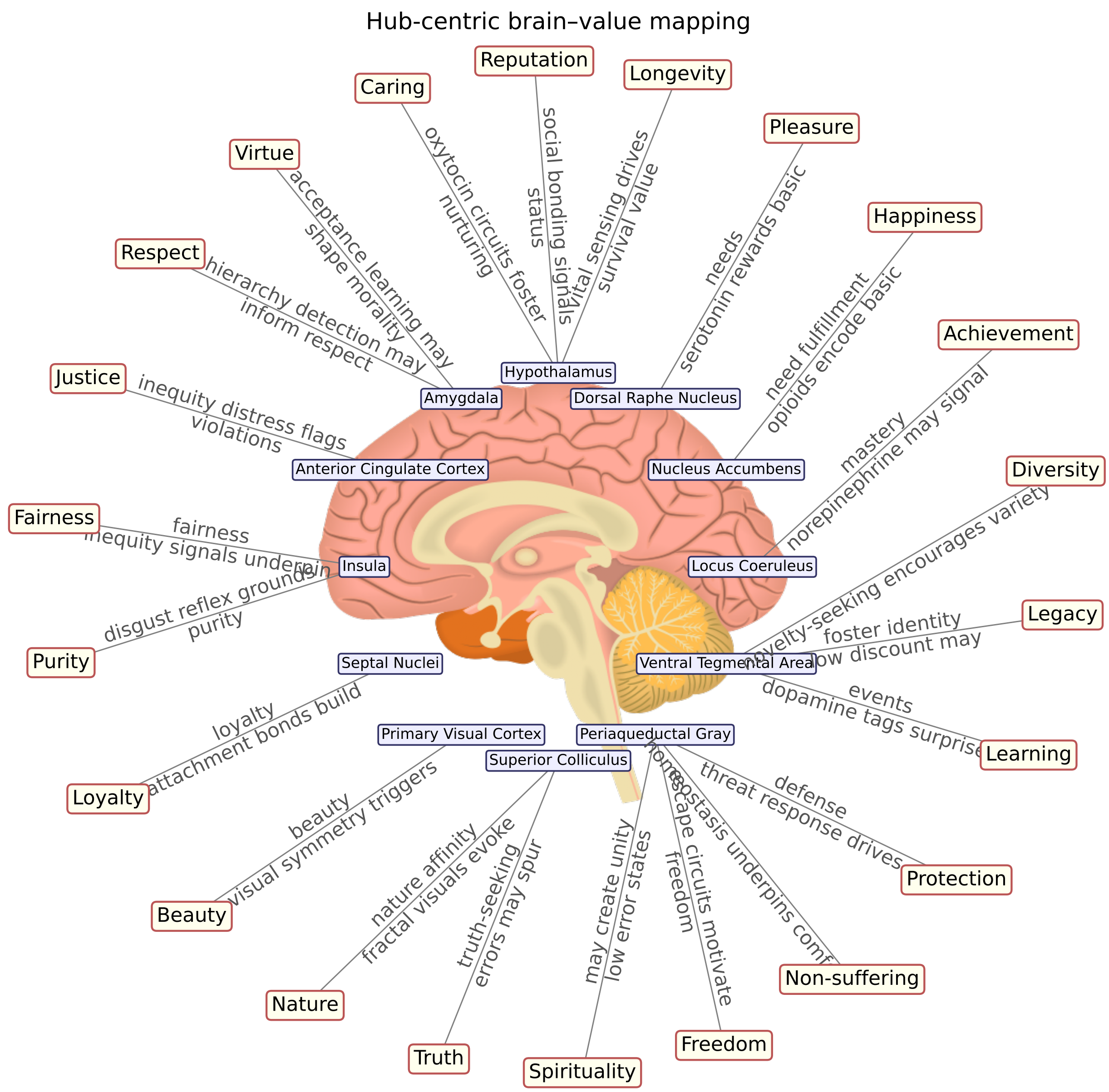

Figure 1: Hub-centric brain–value mapping. Neuromodulatory and salience hubs (inner ring) project intrinsic value tags (outer ring), compiled from Zarncke [1].